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# Edwin Daniel Navarro Monserrat

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## Education:

Ph.D. Plant Pathology – The Ohio State University ***(2019 - 2026)***
MS Plant Pathology – The Ohio State University ***(2019 - 2023)***
BS Cell and Molecular Biology – Universidad de Puerto Rico, Rio Piedras (UPR-RP) ***(2013 - 2019)***

## Experience:

**Bioinformatics Specialist** – Puerto Rico Public Health Lab ***(2025 - Present)***

- Implemented bioinformatic pipelines for analysis of bacterial pathogens, SARS-CoV-2, influenza, and *C. auris* sequences, implementing modules for QC, genome assembly, variant detection, lineage assignment, and AMR profiling.
- Prepared automated reports, dashboards, and downstream analysis of biological and non-biological data to optimize data interpretation related to pathogen surveillance.
- Prepared training materials and authored SOPs for the training of staff members and epidemiologists to enhance reproducibility and foster cross-training initiatives.
- Collaborated cross-functionally with epidemiology and wet-lab teams to strengthen statewide genomic surveillance programs and streamline outbreak response.
- Designed an internal LIMS system to automate sample tracking, sequence data integration, instrument logging, and metrics reporting for improved operational efficiency and data traceability.

**Graduate Research Associate** – The Ohio State University Department of Plant Pathology
*Dr. Taylor Lab - Root-Biotic Interactions Lab* ***(2019 - 2025)***

- Screened a collection of *Pseudomonas* spp. for biocontrol activity against *Pythium* for control of root rot in hydroponically grown leafy greens using in-lab/in vitro testing and greenhouse/hydroponic trials.
- Employed bioinformatic approaches to identify various potential modes of action, including putative secondary metabolites, secretion system-related effectors, and volatiles in a collection of *Pseudomonas* spp.
- Wrote and successfully secured grants to support research projects (7000 USD)
- Presented research outcomes at various conferences and meetings.

**Research Assistant** – The Ohio State University Department of Plant Pathology
*I-CORPS@Ohio, Entrepreneur Lead* ***(April 2019 - July 2019)***

- Market research of a business model centered around the commercialization of university-owned microbes and/or natural products.
- Conducted interviews with over 100 relevant stakeholders across universities, the biotechnological industry, and organizations that also work towards the commercialization of university material.

**Undergraduate Research Assistant –** Universidad de Puerto Rico, Rio Piedras
*Dr. Bayman Lab, Department of Biology (**August 2016 - August 2018**)*

- Performed DNA extractions on a substantial collection of *Pseudocercospora griseola* isolates.

- Executed PCR amplification of four conserved genes, utilizing multi-locus sequencing analysis for genetic characterization.
- Contributed to pathogenicity trials, playing a key role in assessing the virulence of isolates against differential bean lines to determine their races.

**Summer Research Opportunities Scholar** – The Ohio State University
*Dr. Ujor Lab, Department of Animal Science* ***(May 2016 - Aug 2016)***

- Assessed the viability of novel strains of *Clostridium beijerinckii* through ribonuclease P-mediated knockdown of a targeted gene.
- Conducted extensive anaerobic fermentation studies and utilized gas chromatography to quantify butanol production.

*Dr. Taylor Lab, Department of Plant Pathology* ***(May 2015 - Aug 2015)***

- Performed in vitro and in planta assays to determine the efficacy of *Pseudomonas* spp. strains as biocontrol agents against *Agrobacterium rhizogenes*.

**Undergraduate Research Assistant** – Universidad de Puerto Rico, Rio Piedras
*Dr. Tinoco Lab, Department of Chemistry* ***(August 2014 - May 2014)***

- Troubleshooted various methods for protein crystallization of Titanium (IV) bound serum-transferrin. Specifically vapor diffusion and microdialysis methods were tested and optimized.

## Professional Memberships:

- Association of Public Health Laboratories Member ***(June 2025 - Present)***
- Sociedad de Microbiólogos de Puerto Rico ***(August 2024 - Present)***
- Microbiology Society Member ***(March 2024 – August 2025)***
- SACNAS member ***(October 2021 – August 2025)***
- American Phytopathological Society ***(August 2019 - Present)***
- Golden Key International Honour Society ***(May 2017 - May 2019)***
- American Society for Biochemistry and Molecular Biology (ASBMB) ***(August 2016 - May 2018)***

## Leadership experience:

- Active member of the Plant Pathology Graduate Student Association (PPGSA) at The Ohio State University ***(August 2019 - Present)***
- **Roles served as part of PPGSA:**
    - Dept. Graduate Studies Committee PPGSA Representative ***(August 2024 – August 2025)***
    - Dept. Vision Committee PPGSA Representative ***(August 2023 - August 2024)***
    - Plant Pathology Search Committee (for department chair position) PPGSA Representative ***(April 2023 - May 2023)***
    - Social Chair ***(August 2022 - May 2023)***
    - Spring Symposium Chair ***(Aug 2022 - May 2023)***
    - President ***(May 2021 - May 2022)***
    - Academic Affairs Committee PPGSA Representative ***(August 2020 - May 2021)***
    - Student Exchange Seminar Co-chair ***(August 2020 - May 2021)***
    - Plant Sale Co-chair ***(August 2020 - May 2021)***
    - Student Exchange Seminar Committee Member ***(August 2019 - May 2020)***

- Served as the Plant Pathology Dept. Representative at the Wooster Campus for the College of Food, Agricultural, and Environmental Sciences (CFAES) Graduate Student Advisory Committee ***(October 2020 - August 2022)***
- Served as Outreach Coordinator for the American Society for Biochemistry and Molecular Biology UPR-RP Chapter ***(August 2016 - May 2018)***

## Teaching, outreach, communication, and service to the community:

**Led and contributed to various expositions** – The Ohio State University

- OSU Fall Student Involvement Fair: Plant Pathology Exposition ***(August 2024)***
- "We Grow Scientist" outreach event: Plant Pathology Exposition ***(May 2024)***
- CFAES Celebration of Nations: Puerto Rico Exposition ***(April 2022, April 2023, April 2024)***

**Implemented, organized and/or assisted in various dynamics to benefit the career development of my peers** – The Ohio State University

- Led the "Mentoring-Up Discussion Workshop" at the 2023 PPGSA Spring Symposium ***(May 2023)***
- Organized the 2023 PPGSA Spring Symposium ***(May 2023)***
- Organized Career Paths Panel for plant pathology graduate students ***(June 2022)***
- Organized and implemented the PPGSA Buddy System Initiative ***(May 2021)***
- Poster Judge for OSU Plant Sciences Symposium ***(April 2021)***

**Prepared materials, and illustrations and gave talks explaining key concepts and practices in the fields of agriculture, microbiology, and bioinformatics** – The Ohio State University

- Gave a presentation for CienciaPR (directed at K-12 audience). Presentation titled "Aplicando la rama de la bioinformática en escenarios agrícola" ***(September 2023)***
- Gave a presentation for CienciaPR (directed at K-12 audience). Presentation titled: "El uso de microorganismos beneficiosos en la agricultura" ***(November 2022)***
- Gave a presentation (directed to undergraduates) for the Sociedad Estudiantil de Microbiología Industrial at the University of Puerto Rico, Mayaguez Campus ***(February 2023)***

**Prepared slides, lab materials and assisted in experiential learning experiences** – The Ohio State University

- Master Gardeners Training: Introduction to Plant Pathology Lecture ***(March 2022)***
- Phytobacteriology Lab Teaching Assistant ***(August 2021- October 2021)***

**Led and organized experiential learning experiences for K-12 students in low-income communities** – Universidad de Puerto Rico, Rio Piedras

- **Outreach Coordinator** ASBMB-UPRRP (15+ schools impacted) ***(May 2017- May 2018)***

## Mentoring:

- Cientifico Latino GSMI Mock Interviewer ***(January 2025)***
- PPGSA Buddy System – Senior Buddy ***(August 2022- May 2023)***
- PPGSA Buddy System – Senior Buddy ***(August 2021- May 2022)***

- SACNAS MAS Program – Mentee ***(June 2024 – September 2024)***

## Oral Presentations:

- Knowing thyself: Understanding the biocontrol arsenal of a collection of *Pseudomonas* by genomic and phenotypic characterization – The Ohio State University, Dept. of Plant Pathology Seminar ***(November 2024)***
- Genome-wide analysis of biocontrol *Pseudomonas* for the identification of Type VI secretion system-related elements – ICPBB and Biocontrol 2024 Conference (Lightning talk) ***(July 2024)***
- El uso de una colección de *Pseudomonas* como agentes de control biológico – Instituto de Genética Barbara McClintock (Virtual) ***(December 2023)***
- Knowing thyself: Understanding the biocontrol potential of a collection of *Pseudomonas* by genomic and phenotypic characterization - University of Minnesota, Dept. of Plant Pathology Seminar ***(October 2023)***
- Examining the biocontrol potential of a collection of *Pseudomonas* spp. against *Pythium* spp.- Spring Symposium- The Ohio State University, Department of Plant Pathology ***(May 2020)***
- Team 13: Natural Products and Microorganisms at OSU – ICORPS@Ohio NSF Program ***(July 2019)***
- Downregulation of DNA integrity scanning protein A (DisA) for enhanced butanol production in *Clostridium beijerinckii*- Summer Research Opportunities Program – OSU ***(July 2017)***
- Testing the Efficacy of *Pseudomonas* spp. as a Biological Control Agent against *Agrobacterium*- Summer Research Opportunities Program – The Ohio State University ***(July 2016)***

## Poster Presentations:

- Genome-wide analysis of biocontrol Pseudomonas for the identification of Type VI secretion system-related elements – ICPBB and Biocontrol 2024 Conference ***(July 2024)***
- Identification of type VI secretion systems, a potential key for *Pseudomonas* success in biocontrol – OSU Plant Sciences Symposium ***(March 2024)***
- Uncovering T6SSs in a collection *Pseudomonas* spp. – PPGSA Spring Symposium ***(May 2023)***
- *In vitro* testing of *Pseudomonas* for antagonism against *Pythium aphanidermatum* and assessment of plant growth promotion on hydroponically grown spinach -- APS Annual Meeting ***(August 2021)***
- Evaluation of a *Pseudomonas* spp. collection for antagonistic potential against *Pythium* spp. and generation of a consortia – APS Annual Meeting ***(August 2020)***
- Evaluation of a *Pseudomonas* spp. collection for antagonistic potential against *Pythium* spp. and generation of a consortia – OSU Plant Sciences Symposium ***(July 2020)***
- Scientific Community Outreach: Promoting Science to Students at the K-12 Level and STEM Students in ASBMB UPR-RP – Experimental Biology Annual Meeting ***(April 2018)***
- Downregulation of DNA integrity scanning protein A (DisA) for enhanced butanol production in *Clostridium beijerinckii.* Summer Research Opportunities Program – OSU ***(July 2017)***
- Identification of *Pseudocercospora griseola* races in Puerto Rico. XVIII Annual Mycology Congress – Mycology Society of Puerto Rico ***(March 2017)***
- Testing the Efficacy of *Pseudomonas* spp. as a Biological Control Agent against *Agrobacterium*. Summer Research Opportunities Program – The Ohio State University ***(July 2016)***

## Funding, Honors and Awards:

- 2026 APHL Bioinformatics Leadership Program ***(May 2026 - Aug 2026)***

- PPGSA Patricia Ngwira Travel Award (500.00 USD) and supplemental award (250.00 USD) ***(March 2024)***
- Programming funds for PPGSA Spring Symposium (1,402.00 USD) ***(April 2023)***
- OSU-IDI Trainee Transformative Research Grant Award-Co applicant (2,000.00 USD) ***(February 2023)***
- CFAES Internal Grants Program (4,891.00 USD) ***(March 2021)***
- PPGSA Patricia Ngwira Travel Award (266.00 USD) ***(June 2021)***
- Patrick S. Osmer Fellowship, OSU ***(2019, 2025)***
- SROP Scholar, The Ohio State University ***(May 2017 - July 2017)***
- SROP Scholar, The Ohio State University ***(May 2016 - July 2016)***

## Publications:

- **Navarro-Monserrat, E.D**.; Taylor, C.G. T6SS: A Key to Pseudomonas's Success in Biocontrol? *Microorganisms* **2023**, *11*, 2718.
- Serrato-Diaz, L.M., **Navarro-Monserrat, E.D.,** Rosas, J.C. *et al.* Phylogeny of *Pseudocercospora griseola* from Puerto Rico, Central America and Tanzania confirms the existence of an Afro-Andean clade. *Eur J Plant Pathol* **157**, 533–547 **(2020)**
- Serrato-Diaz, L. M.; Rivera-Vargas, L. I.; Goenaga, R.; **Navarro, E. D.;** French-Monar, R. D. First Report of Colletotrichum Fructicola and C. Queenslandicum Causing Fruit Rot of Rambutan (Nephelium Lappaceum). *Plant Disease* **2017**, *101* (6), 1043.

## Technical Skills:

**Computational & Bioinformatics**

- Pathogen genomic surveillance pipeline development and analysis (QC, assembly, variant calling, lineage assignment, AMR profiling)
- Comparative genomics, phylogenetics, and whole-genome sequencing analysis
- Genomic analysis for outbreak tracking

**Programming & Data Analysis**

- R (advanced) and Bioconductor
- Unix/Linux, HPC environments, and bash scripting
- PostgreSQL relational database design
- Data visualization with R (e.g., ggplot2 and RShiny)
- Workflow automation
- Reproducible reporting (Quarto)

**Microbiology & Molecular Biology**

- Bacterial transformation, nucleic acid extractions, PCR, RT-PCR, and electrophoresis
- Experience with bacteria, fungi, oomycetes, and viruses
- Experience with wet lab NGS procedures (e.g., library preparation for Illumina platforms)

**Plant Systems & Applied Microbiology**

- Greenhouse, soil, and hydroponic systems

- Application of beneficial microbes
- Phenotypic screening for plant growth promotion and biocontrol

**Professional Competencies**

- Experimental design and statistical analysis
- Interdisciplinary collaboration
- Science communication to diverse audiences
- Bilingual (Spanish/English)

**References available upon request**